KUMARAGURU COLLEGE OF TECHNOLOGY

24ADT015-FINANCE, ECONOMICS AND MARKETING

NAME : NISHANTH P

ROLL NO : 24BAD405

1. Quantitative Demand Forecasting

Definition:

1. Quantitative demand forecasting involves using **numerical and statistical data** to predict future demand.
2. It depends on **historical sales, market trends, and measurable variables**.
3. The process is **objective**, using mathematical and computational models.
4. Techniques include **time series analysis, regression, and econometric models**.
5. It provides **accurate, data-driven forecasts** for production and inventory decisions.
6. Helps management to **plan pricing, marketing, and distribution** strategies efficiently.
7. It reduces uncertainty and supports **budgeting and resource allocation**.
8. Commonly used in **industries like FMCG, automobiles, and electronics**.

Example (Scenario): A mobile company analyses the last 5 years' sales and advertisement spending to forecast sales for a new model using regression analysis.

2. Time Series Analysis

Definition:

1. A **time series** is data collected at **regular time intervals** (e.g., monthly sales).
2. It studies **patterns and trends** in past data to predict future demand.
3. The main components are **trend, seasonal variation, cyclical variation, and random factors**.
4. **Trend** shows the long-term direction (growth or decline).
5. **Seasonal patterns** repeat at regular intervals, such as festive sales peaks.
6. Used for **short-term forecasting and inventory planning**.
7. Methods include **moving average and exponential smoothing**.
8. It helps in identifying **periodic demand fluctuations**.

Example (Scenario): A clothing retailer uses time series data to predict higher winter jacket sales from November to January every year.

3. Causal Model

Definition:

1. A causal model explains the **relationship between demand and influencing factors**.
2. These factors can include **price, income, advertising, population**, and others.
3. It uses **regression and econometric equations** to find how each factor affects demand.
4. Assumes that changes in independent variables **cause** changes in demand.
5. Provides deeper insight into **market behavior and responsiveness**.
6. Useful for **policy formulation and pricing decisions**.
7. Helps managers test **“what-if” scenarios** to predict demand outcomes.
8. It improves strategic decision-making by identifying **key demand drivers**.

Example (Scenario): A car manufacturer finds that a 5% drop in price increases demand by 8%, showing a strong causal link between price and demand.

4. Delphi Method (Qualitative Forecasting)

Definition:

1. The Delphi method is a **qualitative forecasting technique** using expert opinions.
2. Experts are chosen from **relevant fields** like industry, marketing, and economics.
3. Information is collected through **multiple rounds of anonymous questionnaires**.
4. After each round, a summary is shared with experts for **revision of views**.
5. The goal is to reach a **consensus without bias or influence**.
6. It is most useful when **historical data is unavailable**.
7. Encourages **independent thinking** and avoids dominance in group discussions.
8. Commonly used in **technology forecasting and product innovation**.

Example (Scenario): Before launching an electric scooter, a company asks 15 industry experts to forecast the next 5 years' EV market growth using the Delphi method.

5. Distinction Between Manager Goals and Shareholder Value in the Service Industry

Definition:

1. **Managers' goals** often focus on operational success and internal performance.
2. **Shareholders' goals** are to maximize profit and firm value.
3. Managers may prioritize **growth, service quality, or employee welfare**.
4. Shareholders emphasize **dividends, returns, and market capitalization**.
5. Differences arise due to **time horizon and risk perception**.
6. Managers may reinvest profits, reducing short-term dividends.
7. Alignment is achieved through **performance incentives or governance**.
8. Balanced decisions ensure **long-term sustainability and shareholder satisfaction**.

Example (Scenario): A hotel manager focuses on training staff to improve service quality, while shareholders are more interested in immediate profit margins.

6. Consumer Surplus Effect When RBI Reduces the Interest Rate

Explanation: When the **Reserve Bank of India (RBI)** reduces interest rates, banks offer loans at cheaper rates. This makes it easier for consumers to borrow money for durable goods like houses, cars, or electronics. As borrowing becomes more affordable, **consumers' purchasing power increases**, and they can buy more products at the same or even lower cost.

Consumer surplus is defined as the **difference between what a consumer is willing to pay and what they actually pay**. With reduced interest rates, the total cost of purchase (including interest payments) falls. Hence, consumers derive **greater satisfaction or surplus** from each transaction because they spend less than they were originally willing to.

For example, if a consumer was prepared to pay ₹12 lakh for a car but due to a reduction in the loan interest rate, the effective cost becomes ₹11 lakh, the ₹1 lakh saved becomes their **additional consumer surplus**. This surplus leads to increased consumption, higher demand, and overall economic growth. Lower interest rates also encourage investment, benefiting both consumers and producers.

7. Elastic and Inelastic Demand During War Between Nations

Explanation: During wartime, the **elasticity of demand** for goods and services changes drastically due to uncertainty, scarcity, and shifting priorities. **Elastic demand** refers to goods where a small change in price leads to a large change in demand, while **inelastic demand** refers to goods whose demand remains stable even when prices fluctuate.

In times of war, **essential goods** such as food, fuel, and medicine become **inelastic** because people must buy them regardless of price increases. Even if fuel prices double, citizens and the military still need it for transportation and defense.

On the other hand, **luxury or non-essential items** such as cosmetics, jewelry, or entertainment services show **elastic demand**. As income and purchasing power decline, consumers cut back on these expenses.

For instance, during a war between nations, the demand for fuel and food remains strong despite price hikes, but smartphone or car sales drop significantly. Thus, war shifts consumer focus toward survival and necessities, creating a clear distinction between elastic and inelastic goods.

8. New Mobile Launch – Qualitative & Quantitative Demand Estimation

Explanation: When a mobile manufacturing company plans to launch a new model, it must **estimate demand** using both **qualitative** and **quantitative methods**. Accurate forecasting ensures efficient production, pricing, and marketing decisions.

Quantitative methods use numerical data such as past sales, market trends, income levels, and advertising expenditure. The manager may apply **time series analysis or regression models** to forecast the number of units likely to sell in the upcoming months. For example, based on the previous year's pattern, the company predicts 1 lakh units in the first quarter.

Qualitative methods rely on expert opinion and consumer insight. The manager may use the **Delphi method or market survey** to gather information from distributors, retailers, and potential customers about expected demand.

By combining both approaches, the manager gains a **balanced forecast** — data-driven accuracy from quantitative analysis and human judgment from qualitative methods. This ensures that the company neither overproduces nor faces stock shortages after launch.